

Methods

Forward Euler

$$u_{n+1} = u_n + h f(u_n)$$

Backward Euler

$$u_{n+1} = u_n + h f(u_{n+1})$$

Implicit Midpoint

$$u_{n+1} = u_n + h f\left(\frac{u_n + u_{n+1}}{2}\right)$$

Problems

Linear oscillator (Harmonic oscillator)

$$u_1' = u_2$$

$$u_2' = -u_1$$

Nonlinear oscillator

$$u_1' = u_2 / \|u\|^2$$

$$u_2' = -u_1 / \|u\|^2$$

Pendulum

$$u_1' = \sin(u_2)$$

$$u_2' = -u_1$$

Homework

① Determine the correct "energy" for the pendulum and prove that it's conserved. Implement it in the notebook and test whether it is conserved numerically.

② For any linear ODE system: $u' = Lu$, ($u \in \mathbb{R}^5$)
application of a Runge-Kutta method with m stages results in

$$u_{n+1} = R(hL)u_n$$

Where $R(z)$ is the stability function:

$$R(z) = 1 + z \underline{b}^T (I - \underline{A}z)^{-1} \underline{e}.$$

Here $\underline{b} = \underline{b} \otimes I_s$, $\underline{A} = \underline{A} \otimes I_s$, $\underline{e} = [1, 1, \dots, 1]^T \in \mathbb{R}^m$
and $\underline{e} = \underline{e} \otimes I_s$.

Here " $\otimes$ " is the Kronecker product.

For each of the following methods, construct $R(z)$ for the harmonic oscillator. Predict the behavior of the numerical energy $\|u_n\|_2$ based on the L_2 norm of $R(z)$:

- Explicit Euler
- Implicit Euler
- Implicit midpoint

③ Use Bseries.jl to determine the modified equation for one of these methods applied to one of the above problems. Based on this, predict the behavior of the energy.